

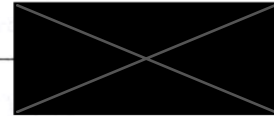
National University of Computer and Emerging Sciences, Lahore Campus



Course: Analog and Digital Communications
Program: BS(Electrical Engineering)
Duration: 3 Hours
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Section: ALL
Exam: Final

Course Code: EE323
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Total Marks: 100
Weight: 50%
Page(s): 2
Roll No:

Instruction/Notes: Attempt all parts of a question together.



Question No. 1

Marks [5+15]

- a) Derive that the signal energy E_g found from the signal $g(t)$ can also be determined from its Fourier transform $G(\omega)$ through Parseval's theorem.
- b) Check whether the given signal $x(t)$ is energy or power signal, find the corresponding energy/power.

$$x(t) = (e^{-5t} + 1)u(t)$$

Question No. 2

Marks [5+15]

- a) If the Fourier Transform of a signal is enough to determine its Spectral Density, why do we complicate our lives by defining the autocorrelation functions?
- b) The random binary signal $x(t)$ shown in Figure 1 transmits one digit every T_b seconds. A binary 1 is transmitted by a pulse $p(t)$ of width $T_b/2$ and amplitude B; a binary 0 is transmitted by no pulse. The digits 1 and 0 are equally likely and occur randomly. Determine the autocorrelation function $\mathcal{R}_x(\tau)$ and the PSD $S_x(\omega)$

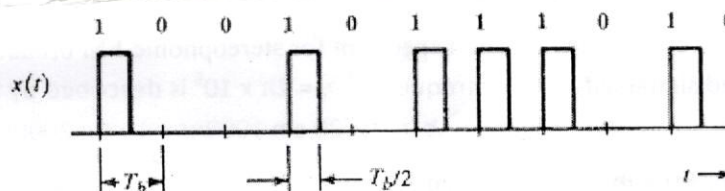


Figure 1

$\frac{1}{2} p(t)$

Question No. 3

Marks [5+15]

- a) The Fourier series for the square pulses $w(t)$ in Figure 2 (a) and $w_o(t)$ in Figure 2 (b) are as follows:

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

$$w_o(t) = \frac{4}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

Show working of a series-bridge diode modulator to accomplish DSB-SC with the help of its block diagram.



Figure 2 (a)

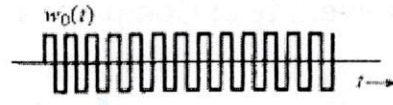


Figure 2(b)

- b) Consider a VSB amplitude modulation system. The baseband signal is an audio signal of bandwidth 4kHz. The carrier frequency $f_c = 1500\text{kHz}$. Suppose the transmission vestigial filter $H_i(\omega)$ has a frequency response as shown in Figure 3 below

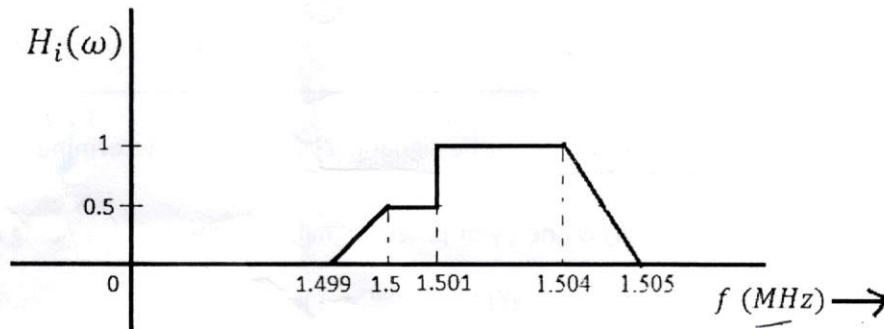


Figure 3

- 1) Design and illustrate a receiver system block diagram (synchronous detector).
- 2) Find the bandwidth of this transmission.
- 3) Describe and sketch the necessary equalizer filter response $H_o(\omega)$ for distortion-less reception.

Question No. 4

Marks [5+15]

- a) Draw and explain the composite baseband spectrum for stereophonic FM broadcasting.
- b) An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 10^6$ is described by the equation
- $$\varphi_{EM}(t) = 5 \cos(\omega_c t + 20 \sin 1000\pi t + 10 \sin 2000\pi t)$$

- 1) Find the power of the modulated signal.
- 2) Find the frequency deviation Δf and the phase deviation $\Delta\varphi$.
- 3) Estimate the bandwidth of $\varphi_{EM}(t)$.

Question No. 5

Marks [5+15]

- a) Show that a Delta demodulator is just an accumulator, assuming zero initial condition $m_q[0]=0$.
- b) Five sensor output signals, each of bandwidth 240 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 30% above the Nyquist rate. Framing and synchronization requires an additional 1.5% extra bits. The PCM encoder uses a non-uniform quantizer with $\mu = 100$ to convert these signals before they are time-multiplexed into a single data stream. The required SNR is to be at least 43dB in this case.
- 1) Find the minimum number of bits required to code the non-uniform quantizer.
 - 2) Determine the minimum possible data rate (bits per second) that must be transmitted
 - 3) Determine the minimum bandwidth required to transmit the multiplexed signal.